

**"RESPONSIVE WEB-BASED PATIENT REFERRAL AND ADMISSION SYSTEM
FOR SALAAM HOSPITAL"**

COLLEGE OF INFORMATION AND COMPUTING STUDIES
Mindanao State University – Main Campus

A Capstone Project 1
Presented to the Faculty of
College of Information and Computing Sciences

In Partial Fulfillment of the Requirements for the Degree of
Bachelor of Science in Information Technology

By:
Abdul Naafi A. Macapodi
Alimar B. Datunaim Al-raafi
C. Indanan

January 2025

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Chapter 1

Introduction

The Hospital management involves the overall administration and coordination of hospital operations to ensure the efficient delivery of healthcare services, including the management of staff, facilities, patient records, and operational workflows. One important component of hospital management is the admission process, which focuses on registering patients, handling referrals, assigning appropriate departments or wards, and ensuring accurate documentation of patient information. Effective management of admission procedures is essential to reduce delays, minimize errors, and improve patient flow within the hospital. Studies have shown that standardizing and improving hospital admission and discharge processes can significantly enhance operational efficiency and reduce patient waiting time, thereby improving overall hospital performance and patient outcomes (Ortiga et al., 2012).

Salaam Hospital, located in Blis Papandayan, Marawi City, currently operates without a formal digital system for managing patient records, admissions, and referrals. Because the hospital's processes are largely manual, essential tasks such as tracking patient history, identifying which doctor or nurse assisted a patient, recording administered medications, and managing patient admissions are time-consuming and prone to human error. The lack of a central system hinders staff efficiency and negatively impacts patient care; therefore, a comprehensive hospital management and admission system is critically needed to support daily operations and improve the quality of services provided to patients.

The absence of an effective system at Salaam Hospital has resulted in several operational challenges. With manual documentation, staff struggle to retrieve accurate historical patient information quickly, which can delay treatment decisions and complicate referrals among in-house specialists. In traditional paper-based record systems, healthcare providers often spend significant time searching for misfiled or missing patient folders, which negatively affects the timely delivery of healthcare services and the standard of care (Influence of Health Information Management Practices on the Quality of Healthcare, 2023).

This study focuses on the development of a Hospital Management and Admission System aimed at modernizing hospital operations. The system is designed to handle various aspects of patient care, including medical history tracking, specialist referrals, treatment and medication documentation, and admission processes. By digitizing records and automating key functions, the system provides hospital staff with timely access to accurate patient information, facilitates communication among healthcare providers, and streamlines administrative tasks. Hospital Information Systems (HIS) have been shown to improve clinical workflows, reduce administrative delays, and enhance the quality of care by supporting efficient decisionmaking and effective management of patient data (Mohammadpour et al., 2021; Influence of Health Information Management Practices on the Quality of Healthcare, 2023).

1.1 Project Context

This capstone project entitled “Responsive Web-Based Patient Referral and Admission System for Salaam Hospital” focuses on addressing the inefficiencies in the hospital’s current manual processes for patient admission, medical record tracking, and internal specialist referrals. At present, activities such as recording patient information, managing admission details, tracking medical history, and coordinating referrals among specialist doctors are conducted using handwritten forms, logbooks, or verbal communication. These manual processes are often time-consuming and prone to human error, leading to misplaced or incomplete records, delayed decision-making, miscommunication between staff, and difficulties in retrieving accurate patient histories. Such inefficiencies not only hinder smooth hospital operations but can also compromise patient safety and the overall quality of care (Shiferaw, Zewdu, & Sisay, 2023; Alemu, Mekonnen, & Getachew, 2023).

To address these challenges, the project proposes the development of a computerized Hospital Management and Admission System that automates patient registration, admission tracking, medical history management, and internal referrals, while providing accessible digital records through both web and mobile platforms. By digitizing records and streamlining workflows, the system is expected to reduce

administrative bottlenecks, minimize errors, and facilitate timely access to critical patient information. The system supports patient admission workflows, including initial triage and internal referrals, particularly for patients requiring urgent or inpatient care. Hospital Information Systems have been demonstrated to significantly improve operational efficiency, enhance interdepartmental communication, support clinical decision-making, and ultimately improve the quality of patient care (Al-Shorbaji, Atun, & Car, 2016; Khosravi, Kazemi, & Hosseini, 2023).

1.2 Purpose and Description

This project entitled “Responsive Web-Based Patient Referral and Admission System for Salaam Hospital” is intended to design and develop an integrated hospital management and admission system that will automate and improve the hospital’s current manual processes of patient admission, medical record management, and in-house specialist referrals. The main purpose of this study is to provide an efficient, accurate, and organized digital system that will support hospital staff in managing patient information, reduce delays in admission and referral processes, and improve overall healthcare service delivery.

This study aims to computerize the hospital’s existing paper-based operations, where patient information, medical history, treatment records, and referrals are manually documented and tracked. The researchers seek to address the inefficiencies, errors, and delays caused by these manual and timeconsuming tasks through the development of a web-based and mobile-based Hospital Management and Admission System. By implementing this system, Salaam Hospital can improve coordination among departments, minimize administrative workload, and ensure that patient records are accurate, complete, and easily accessible.

Through the proposed system, it will enable healthcare providers to access real-time patient information, support better clinical decision-making, and enhance workflow efficiency. Ultimately, the

system is expected to improve the quality of patient care, reduce waiting times, and increase patient satisfaction.

Beneficiaries of the Study

• Hospital Administration

The system will help hospital administrators efficiently monitor admissions, patient records, and internal referrals. It will provide access to accurate and real-time data, enabling better planning, reporting, and decision-making for hospital operations.

• Medical Staff (Doctors and Nurses)

Doctors and nurses will benefit from quick and easy access to patient medical histories, treatment records, and referral details. The system will reduce paperwork, improve coordination among in-house specialists, and help ensure timely and accurate patient care.

• Administrative Staff

The system will simplify admission processes, record management, and documentation tasks. Automating these processes will reduce workload, minimize errors, and speed up daily hospital operations.

• Patients

Patients will experience faster admissions, smoother referrals among specialist doctors, and improved healthcare services. Accurate record-keeping will also ensure better continuity of care and reduced waiting times.

• Researchers

The researchers will gain practical experience in system analysis, design, and development within a healthcare environment. This study will enhance their technical skills and understanding of real-world hospital workflows.

1.3 Objectives

1.3.1 General Objective

This capstone project aims to design and develop a hospital management and admission system that will assist the beneficiaries in their hospital operation.

1.3.2 Specific Objectives

To achieve the general objective of the study, the following specific objectives are identified:

1. To gather data and conduct requirements analysis by assessing the current manual processes of Salaam Hospital, including patient admission, medical record and internal referral workflows.
2. To design the system processes, user interface, and database structure that will be use on the system development.
3. To develop a responsive web application, based on the approved design and requirements.
4. To test and evaluate the developed system through functional testing and usability evaluation to ensure that it meets the required performance, reliability, and usability standards of Salaam Hospital.

1.4 Scope and Limitations

This capstone project focuses on the design and development of a computerized system intended to assist the hospital's patient referral and admission processes. The study covers only the activities related to recording, managing, and monitoring of patient referral and admission information within Salaam Hospital. It also includes the coordination between hospital departments involved in referring and admitting patients. The project is primarily concerned with data management, documentation, and communication related to the referral and admission workflow of the hospital.

The system includes basic medical record management related to patient admission, internal

referral, diagnosis, triage, and treatment documentation but does not support full laboratory systems, billing, or external health system integration.

This study is limited to the administrative and operational aspects of handling patient referrals and admissions. It does not include financial transactions, and billing. External hospital networks outside Salaam Hospital and third-party health institutions are also not covered in this project. The proposed system's focus is on improving internal hospital efficiency in processing referrals and admissions, without addressing broader health information systems or government-level health data integration.

Chapter 2

Review of Related Literature and Systems

This chapter presents the theoretical foundations that underpin the development of the "Responsive Web-Based Patient Referral and Admission System for Salaam Hospital". It organizes the discussion into local, national, and international theories to provide a comprehensive scientific and conceptual basis for the study. By examining these theoretical frameworks, the chapter establishes the relevance, justification, and scholarly grounding of the proposed system within the broader context of healthcare, information technology, and systems development.

2.1 Review of Related Concepts

2.1.1 Hospital Management Systems (HMS)

Hospital Management Systems (HMS) are comprehensive software platforms that automate and streamline the operational, administrative, and clinical processes of healthcare facilities. HMS integrates multiple modules, including patient registration, appointment scheduling, billing, laboratory management, pharmacy, and human resource management. By centralizing all hospital operations, HMS reduces paperwork, prevents duplication of records, and improves communication between departments. The implementation of HMS enhances hospital efficiency by allowing staff to access patient and operational data in real time, enabling informed decision-making and timely response to patient needs. For instance, a patient admitted through the HMS can have their medical history, lab results, and billing details immediately available to doctors, nurses, and administrative staff, which reduces delays and errors. Moreover, HMS supports reporting and analytics functions that help hospital management monitor resource utilization, track performance, and plan improvements in healthcare delivery. This system ultimately contributes to higher quality patient care and smoother hospital workflows.

2.1.2 Patient Admission and Discharge Systems

Patient Admission and Discharge Systems are specialized software solutions that manage the complete lifecycle of a patient's stay in a healthcare facility, from admission to discharge. These systems handle patient intake, bed allocation, recording of medical histories, generation of discharge summaries, and coordination with billing departments. By automating these processes, hospitals can reduce patient waiting times, minimize errors in documentation, and improve workflow efficiency among healthcare staff. According to research, such systems enhance operational management by providing real-time updates on bed availability, patient status, and required treatments. For example, when a patient is admitted, the system can automatically notify nurses, doctors, and lab personnel about the patient's condition and scheduled tests, ensuring timely care. Additionally, centralized recordkeeping ensures that all patient information is securely stored and easily accessible for future reference, supporting continuity of care and informed decisionmaking.

2.1.3 Integration of EMR/EHR with Hospital Management Systems

The integration of Electronic Medical Records (EMR) and Electronic Health Records (EHR) with Hospital Management Systems (HMS) allows hospitals to create a seamless, centralized digital platform that connects administrative and clinical processes. This integration enables real-time synchronization of medical histories, diagnostic reports, treatment plans, and billing information across departments. Studies indicate that such integration improves operational efficiency, reduces redundant testing, and enhances continuity of care by ensuring that all healthcare providers have access to accurate and up-to-date patient information. For example, when a patient is transferred from one department to another, doctors and nurses can immediately access previous test results, medications, and referral notes, minimizing delays and potential errors. Moreover,

integrating EMR/EHR with HMS supports advanced analytics, enabling hospitals to monitor performance, plan resource allocation, and improve service delivery.

2.1.4 Digital Admission Management Systems in Hospital Operations

Digital admission management systems in hospitals provide a centralized electronic platform that streamlines patient intake and replaces traditional manual or paper-based admission workflows. These systems often include features such as online pre-registration, real-time tracking of patient admissions, and integrated data capture at the point of service, allowing administrative and clinical staff to manage admissions efficiently. Research indicates that digital admission systems improve operational efficiency, reduce patient waiting times, and enhance coordination among registration, clinical, and administrative units.

Hospitals implementing such systems report faster patient turnaround, while patients and families benefit from greater convenience and reduced waiting. Electronic admission platforms also minimize administrative workload, decrease documentation errors, and provide real-time access to bed availability, admission progress, and patient flow. For example, when a patient is admitted through the digital system, all relevant staff can immediately access their registration details and medical information, ensuring smooth transitions between departments. Integrating digital admission management into the proposed Hospital Management and Admission System for Salaam Hospital can therefore optimize patient flow, improve data accuracy, and support higher-quality service delivery.

2.2 Review of Related Studies

2.2.1 Design and Development of a Web-Based Patient Management Information System

The study titled “Design and Development of a Web-Based Patient Management Information System” by Shah et al. (2024) explored the implementation of a web-based hospital management system designed to streamline patient registration, queue management, medical record tracking, and automated reminders. The system aimed to enhance operational efficiency by reducing reliance on manual processes and providing centralized access to patient data.

Findings indicated that web-based management improved hospital workflow, minimized delays in patient processing, and reduced administrative errors. This study is directly relevant to the proposed system for Salaam Hospital as it demonstrates the benefits of digitalizing patient admission and record management processes. By adopting a similar web- and mobile-based approach, Salaam Hospital can reduce manual errors, improve coordination among staff, and provide faster access to patient information.

2.2.2 Acceptance Model of a Hospital Information System

The study “Acceptance Model of a Hospital Information System” by Tung et al. (2016) examined factors influencing the adoption of Hospital Information Systems (HIS) among hospital staff, focusing on technological, human, and organizational determinants. Using survey and case study methods, the research found that staff engagement, ease of use, and perceived usefulness significantly impacted HIS acceptance.

This research highlights that user acceptance is as important as technical design. For Salaam Hospital, ensuring that doctors, nurses, and administrative staff find the system easy and beneficial to use will determine its success. The proposed system should prioritize usability and stakeholder involvement to encourage adoption and maximize its operational impact.

2.2.3 Integration of Electronic Medical Records Between Primary and Specialist Care Institutions

The study entitled “The Benefits of Integrating Electronic Medical Record Systems Between Primary and Specialist Care Institutions” analyzed how shared and integrated EMR systems influence patient referrals, medical record tracking, and coordination of care. The findings revealed that EMR integration improves continuity of care by allowing healthcare providers from different departments or specialties to access unified patient information. The study emphasized that centralized medical records reduce redundant testing, minimize information gaps, and enhance collaboration among healthcare professionals.

This study is relevant to the current capstone project because it supports the development of a centralized hospital management system that integrates patient medical records and referral information. By enabling seamless access to patient data across departments, the proposed system improves referral efficiency, enhances coordination between specialist doctors and other healthcare staff, and supports effective hospital service delivery.

2.2.4 Improving the Inpatient Referral System in Acute Medical Care

The article “Improving the inpatient referral system in the acute medical setting” focused on an electronic referral system designed to replace traditional paper-based referral processes for all medical specialties. The findings revealed that the system facilitated faster submission of referrals, better data accuracy, reduced paperwork, and improved shared access to referral information across departments. Users also reported a reduction in errors and enhanced overall workflow as compared to manual referral tracking.

This study supports the rationale for your project by highlighting that electronic referral systems streamline internal hospital referrals — a core component of your proposed system for Salaam

Hospital. By digitizing referral forms and automating communication between departments, your system can achieve similar improvements in accuracy, speed, and coordination of patient care.

2.2.5 Real-Life Implementation and Evaluation of an Electronic Referral System

A study titled “Real Life Implementation and Evaluation of the e-Referral System SIPILINK” examined the deployment and early outcomes of an electronic referral platform designed to streamline intra-hospital referral workflows, enhance communication among clinical staff within the same hospital, and improve patient care pathways. The research found that SIPILINK significantly improved the efficiency of the referral process by enabling faster communication between physicians and specialists, reducing delays in internal care coordination. The system also demonstrated effectiveness in supporting direct admissions within the hospital by facilitating rapid acceptance and processing of internal specialist referrals.

This study is particularly relevant to the proposed Hospital Management and Admission System for Salaam Hospital because it validates the practical advantages of implementing a hospital-based electronic referral system in real clinical settings. The findings demonstrate how digital referral features can improve communication and coordination among departments and healthcare providers within a single hospital, reduce waiting times, minimize miscommunication, and enhance overall operational efficiency in managing internal referrals and admissions.

2.2.6 Lessons from a Web-Based Referral System for Acute Neurosurgical Referrals

The article “Lessons from a Web-Based Referral System for Acute Neurosurgical Referrals. Part 1: System Features and Evaluation” examined the implementation and evaluation of a web-based electronic referral system designed to replace informal paper-based referral processes within a single hospital, particularly among departments referring patients to the neurosurgical unit. The system was developed to support intra-hospital specialist referrals,

enabling physicians from different departments within the same hospital to electronically submit and track neurosurgical referral requests.

The study found that transitioning to an electronic referral platform resulted in more organized and searchable referral records, improved documentation quality, and reduced reliance on incomplete or illegible paper referrals. The system provided a centralized database where each internal referral contained detailed patient information and timestamps, thereby improving communication, traceability, and coordination among hospital departments involved in the referral process.

This study supports the objectives of the proposed Hospital Management and Admission System for Salaam Hospital by demonstrating the effectiveness of a hospital-based digital referral workflow. By centralizing referral data and automating communication within the hospital, the system can reduce documentation errors, improve response times, and enhance coordination among in-house specialists. These outcomes align with the proposed system's goals of streamlining internal referral processes, improving patient management, and increasing overall operational efficiency.

2.3 Review of Related Systems

2.3.1 Integrated Hospital Operations and Management Information System(iHOMIS)

The Integrated Hospital Operations and Management Information System (iHOMIS) was developed by the Department of Health (DOH), Philippines, to modernize hospital operations and optimize healthcare service delivery in public hospitals. Its main objective is to digitally centralize hospital processes such as patient admission, registration, billing, laboratory management, pharmacy coordination, and medical record maintenance. By doing so, the system allows healthcare staff to access real-time patient information, track the status of admissions, and make data-driven decisions efficiently. iHOMIS operates through a centralized database that integrates multiple

modules, including admission and registration which collects patient personal information, assigns beds, and schedules treatments; medical record management which maintains electronic health records including past consultations, lab results, and referral history; billing and PhilHealth claims processing which automates fee computation, payment tracking, and insurance documentation; and reporting and analytics which generates operational and clinical reports to support hospital management and government compliance. The system has been implemented in several DOH public hospitals, improving operational efficiency, reducing human errors caused by manual documentation, and facilitating timely patient care. It is widely used by hospital administrators, doctors, nurses, and clerical staff, making it an essential tool for public hospital operations. This system is highly relevant to the capstone project because it exemplifies the integration of patient admission and history tracking in a centralized platform. The proposed Web and Mobile-Based Hospital Management System builds upon these concepts, providing web and mobile accessibility, real-time updates, and a user-friendly interface for tracking patient records and admissions efficiently.



Figure 2.1: Integrated Hospital Operations and Management Information System(iHOMIS)

2.3.2 CNR Software Systems

The CNR Software Systems developed the Appoluz Hospital Information System (HIS) and Electronic Medical Records (EMR) to automate hospital operations and improve the efficiency of patient care in Philippine health facilities. Its primary purpose is to streamline workflows, including patient registration, admissions, medical records management, laboratory and radiology processes, pharmacy inventory, and billing, thereby reducing manual paperwork and minimizing errors. The system was designed as a comprehensive digital platform that integrates all major hospital departments and supports real-time data management. Patients' medical information is digitized through EMR, allowing doctors and nurses to access historical records, vital signs, diagnoses, and e-prescriptions easily. The system also supports ICD-10 coding, automated billing, and PhilHealth or HMO processing to ensure regulatory compliance and accurate documentation. CNR Software Systems developed the platform locally in the Philippines, focusing on cost-effective solutions accessible to hospitals of various sizes. The Appoluz HIS incorporates features such as smart pharmacy inventory, laboratory and radiology result management, patient queuing, and financial reporting modules. It is DOH-accredited, which ensures that it meets national standards for e-health systems. Many hospitals and clinics in the Philippines, including Clinica Antipolo, Soriano-Leyble Medical and Maternity Hospital, St. Therese Hospital, San Isidro Hospital, and Taytay Maternity and Child Care Center, have implemented the system. Reported benefits include improved workflow efficiency, reduced paperwork, better coordination among departments, accurate and faster billing, and streamlined access to patient information. The system allows hospital staff to monitor patient progress, track laboratory and pharmacy usage, and maintain a centralized database for easier decision-making. While there is no formal peer-reviewed study quantifying its impact, the adoption of Appoluz HIS has reportedly enhanced operational efficiency, improved documentation, and supported hospitals in providing more organized and reliable patient care.

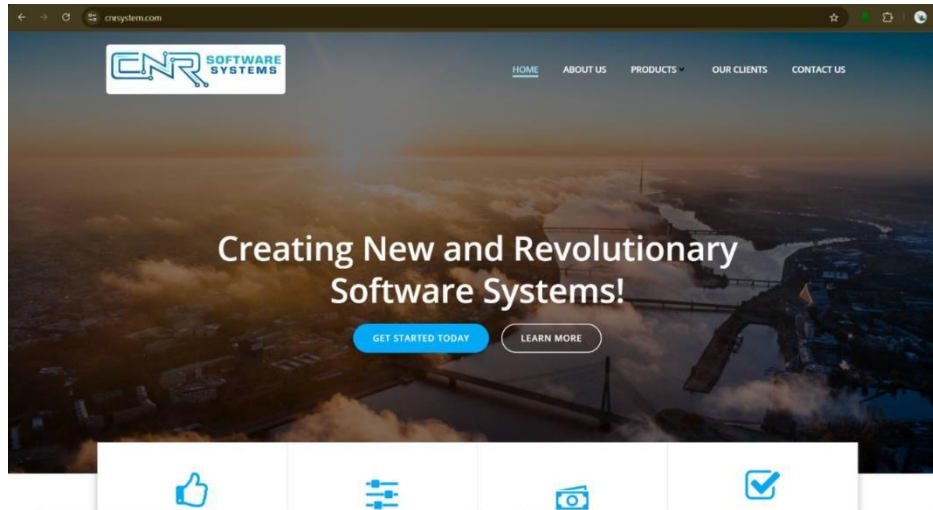


Figure 2.2: CNR Software Systems

2.3.3 WAH (We Are Health)

The We Are Health (WAH) Platform is a Philippine-based digital health solution designed to support health facilities and local government units (LGUs) in managing patient information, clinical workflows, and health program monitoring. Its primary purpose is to streamline hospital and clinic operations, improve record-keeping, and enhance patient care through digital solutions. WAH is composed of several modules, including WAH4C for clinics, WAH4H for hospitals, WAH4P for patients, and WAH4Interoperability for sharing patient information across health facilities. These modules enable electronic medical records, laboratory and radiology result management, pharmacy inventory, patient queuing, automated billing, and health alerts. The system was developed to be cloud-based and accessible, providing real-time data management and integration with national health programs. WAH supports interoperability between multiple health facilities, allowing patient information to be securely shared and coordinated across different clinics and hospitals. Health facility staff such as doctors, nurses, and administrative personnel are the primary users, while patients can access the platform for appointment reminders, health alerts, queuing, and feedback. The platform employs a “Train Now, Pay Later” scheme for LGUs and

facilities with limited budgets, while some patient-facing features are expected to be free of charge. WAH has been reported to reduce manual documentation, improve workflow efficiency, and enhance patient follow-up, particularly for rural health units implementing national health programs. Patients benefit from timely reminders, organized queuing, and better access to their medical history, which can lead to improved clinical outcomes. Overall, the WAH platform demonstrates how an integrated digital health system can optimize hospital and clinic operations, improve patient engagement, and support efficient healthcare delivery in the Philippine context.

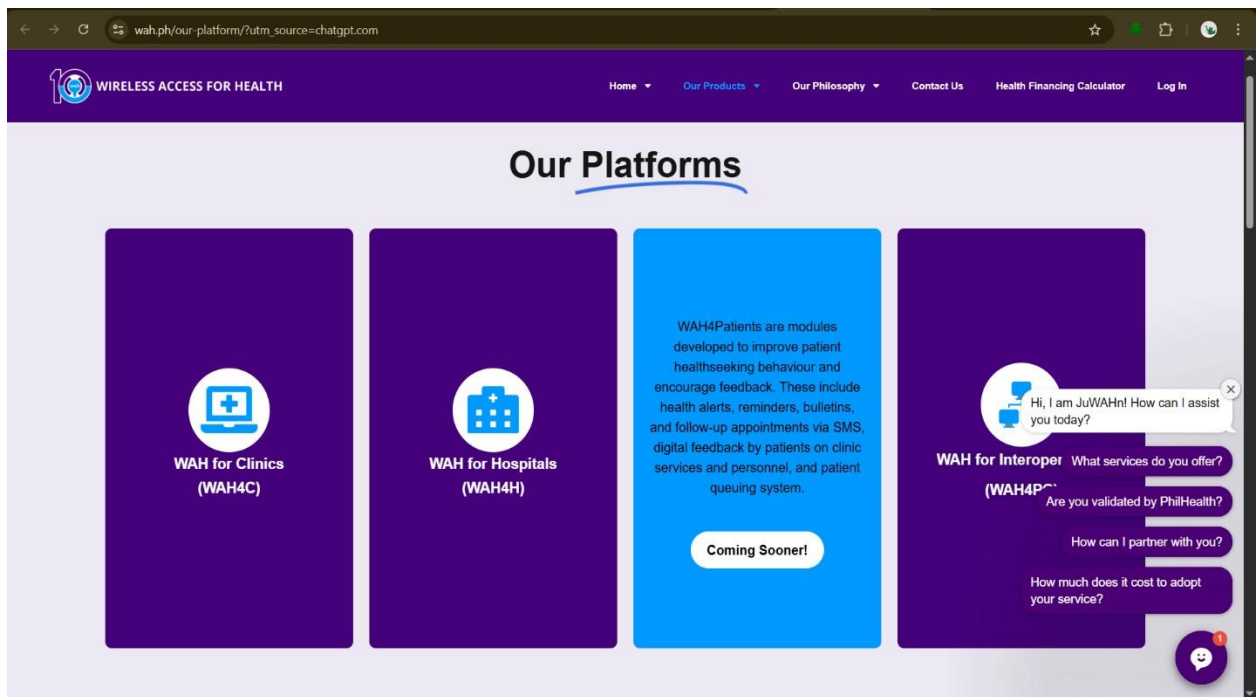


Figure 2.3: WAH (We Are Health)

2.3.4 Phreesia Referral Management System

The Phreesia Referral Management System, developed by Phreesia, Inc. in the United States, streamlines patient referrals among healthcare providers and hospitals. It centralizes referral information, including demographic data, clinical notes, and supporting documents, into a single digital platform. The system can capture referrals sent through traditional methods like fax,

converting them into structured digital records, and integrates with EMR/EHR systems to reduce duplication of patient information.

The platform also automates patient outreach through text or email, tracks referral status, and coordinates appointments, reducing administrative workload and improving patient care continuity. Widely used in hospitals and clinics, Phreesia Referral Management enhances efficiency, referral tracking, and inter-provider coordination.

By providing real-time visibility of referral status and outcomes, the system supports better clinical decision-making and timely patient care. Healthcare organizations using Phreesia have reported faster referral processing, improved coordination between referring and receiving providers, and increased patient satisfaction. Its functionality is particularly relevant to the proposed Mobile and Web-Based Patient Referral and Admission System for Salaam Hospital, serving as a model for automating and centralizing referral workflows.

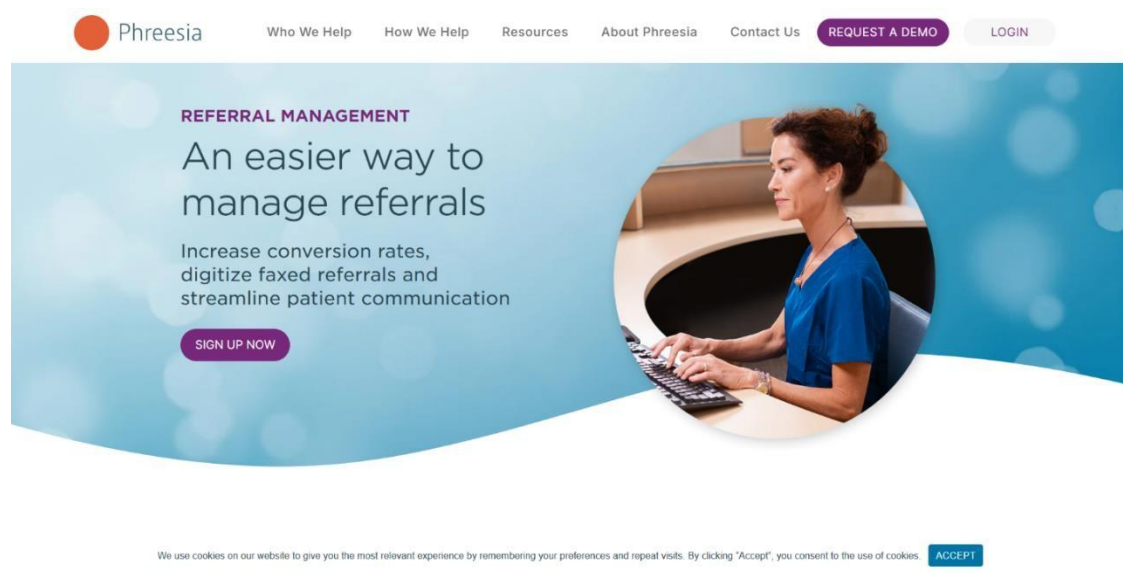


Figure 2.4: Phreesia referral management system

2.3.5 WellSky Enterprise Referral Manager

The Bizbox Hospital Information System (HIS) is a commercially licensed software platform developed by Bizbox Inc., a Philippine IT company, designed for medium to large hospitals and healthcare facilities in the Philippines. Its primary function is to automate and integrate hospital operations, including patient admissions, medical record management, billing, inventory tracking, and staff coordination. Key features of Bizbox HIS include patient history and medical record tracking which allows doctors and nurses to access a patient's complete medical history including prior consultations, medications, lab results, and referrals; admission and discharge management which automates the process of patient intake, bed allocation, monitoring, and discharge documentation; billing and insurance integration which supports accurate fee computation, PhilHealth claim submission, and reporting for financial departments; and resource coordination and workflow optimization which enables staff to monitor real-time hospital operations and plan resource allocation effectively. Bizbox HIS is used by numerous private hospitals, specialty clinics, and healthcare chains across the country, and its implementation has led to improved operational efficiency, reduced errors, better patient data management, and enhanced decision-making for hospital staff. This system is relevant to the capstone project because it demonstrates how a full-scale hospital management system integrates multiple modules that handle admission, patient tracking, and referrals. The proposed Web and Mobile-Based Hospital Management System draws from these concepts, adding enhanced accessibility, mobile interface, and real-time tracking to further improve patient care and hospital workflow efficiency..

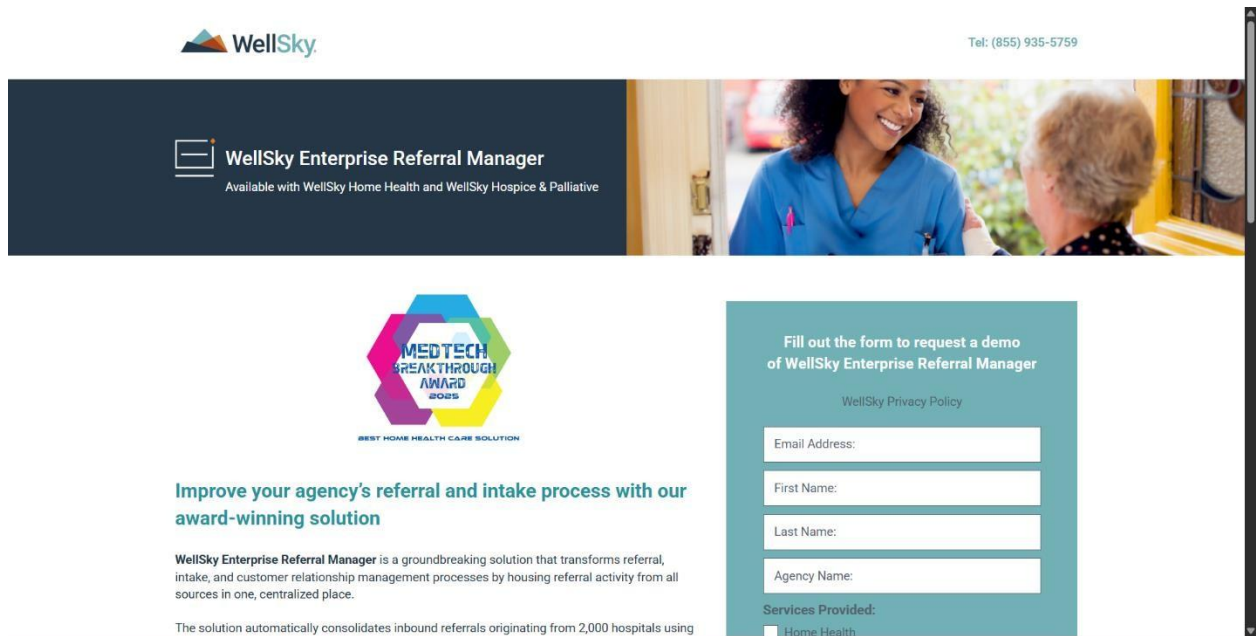


Figure 2.5: WellSky Enterprise Referral Manager

2.3.6 Summary of Related Systems

The review of related systems highlights the growing trend of integrating digital platforms to streamline hospital operations, patient admission, referral management, and clinical workflows. Systems such as iHOMIS, Appoluz HIS, and WAH demonstrate the benefits of centralized digital platforms that combine patient registration, medical record management, laboratory and pharmacy coordination, billing, and reporting, resulting in improved operational efficiency and reduced administrative workload. Meanwhile, referral-focused platforms like the Phreesia Referral Management System and WellSky Enterprise Referral Manager emphasize automation, centralization, and real-time tracking of patient referrals, enhancing communication between departments and providers, minimizing delays, and supporting timely clinical decision-making. Collectively, these systems show that the implementation of web- or mobile-based hospital management solutions leads to better accessibility, accurate and up-to-date patient information, improved workflow efficiency, and enhanced patient care. The proposed "Responsive Web-Based

Patient Referral and Admission System for Salaam Hospital" builds upon these concepts by integrating centralized patient admission, history tracking, and internal referral workflows, while providing real-time updates and mobile accessibility to further optimize hospital operations and service delivery.

Table 2.1: Summary of Related Systems

Key Features	iHOMIS (PAS)	CNR Systems	We Are Health	WellSky	Phreesia	Proposed System
Hospital Workflow Integration						
Real-Time Referral / Admission Tracking						
Multi-Department Coordination						
Web-Based Access						
Mobile Access						
Automated Notifications (In-App)						
User Data Entry (Health Staff)						
Centralized Patient Referral Database						
Role-Based Access Control (Admin, Doctor, Nurse, Admissions)						
Patient Referral Status & History Log						
Priority & Triage Level Tagging						
Document Upload (Referral Forms, Lab Results)						

Reporting & Analytics (Referrals, Admissions, Turnaround Time)						
Data Security & Privacy Controls						

2.4 Technical Background

Salaam Hospital is a key healthcare institution serving the local community, offering medical services, outpatient care, emergency response, and patient admissions. Like many mid-sized hospitals in the Philippines, it caters to a growing population with diverse medical needs. With increasing healthcare demands, efficient management of patient flow, hospital operations, and care provider coordination is critical.

Currently, hospital operations at Salaam Hospital are largely manual and paper-based. Patient registration, admission, and care tracking rely on handwritten forms, phone calls, or verbal communication. Records are often incomplete, illegible, or fragmented across departments, personal devices, and offline logs. This results in delays in patient care, inefficient staff coordination, and difficulty in tracking resource utilization or patient history.

A significant challenge is the absence of a centralized digital platform integrating **patient admission, medical records, and care provider tracking**. Information is scattered, making it difficult to monitor patient status, coordinate staff assignments, and generate accurate reports on hospital performance. Existing digital health systems like PAS, Appoluz HIS, and WAH focus mainly on appointments, electronic medical records, or internal workflows. They lack a unified system that tracks patient admissions while monitoring the activities and assignments of doctors, nurses, and other care providers in real time.

2.4.1 Organizational Chart

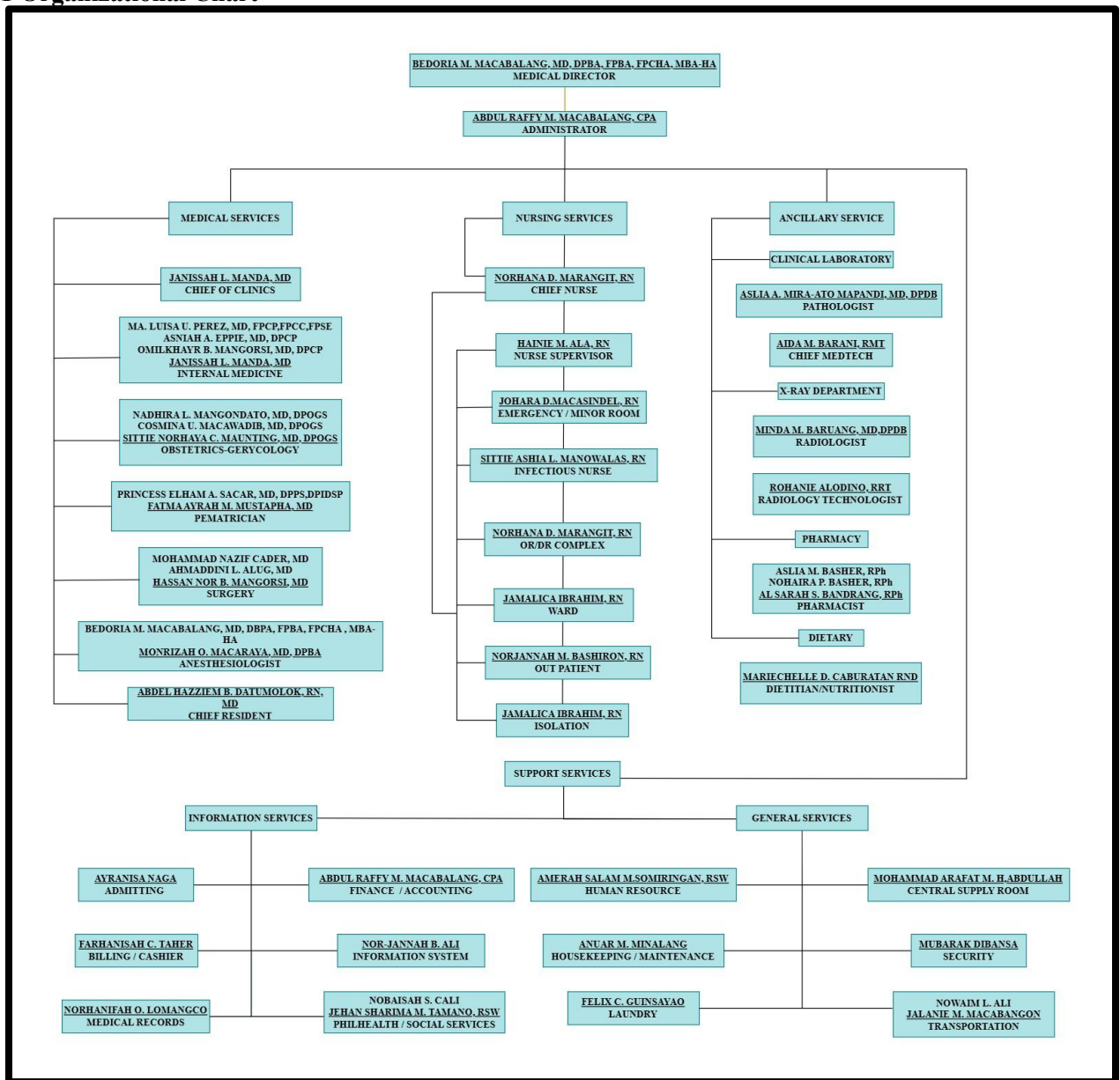


Figure 2.6: Organizational Chart of Salaam Hospital

2.5 Workflow

2.5.1 Current Workflow

Salaam Hospital primarily manages patient admissions, medical records, and internal referrals through manual and paper-based processes. Upon arrival, patients' personal information and admission details are recorded by staff using handwritten forms, logbooks, and physical patient charts.

Doctors and nurses document medical histories, diagnoses, treatments, and medications on paper records. When an internal referral to an in-house specialist is needed, the process is handled through verbal communication or handwritten referral notes. Patient records are stored physically in folders and filing cabinets, requiring staff to manually search and retrieve files whenever information is needed for treatment or reporting.

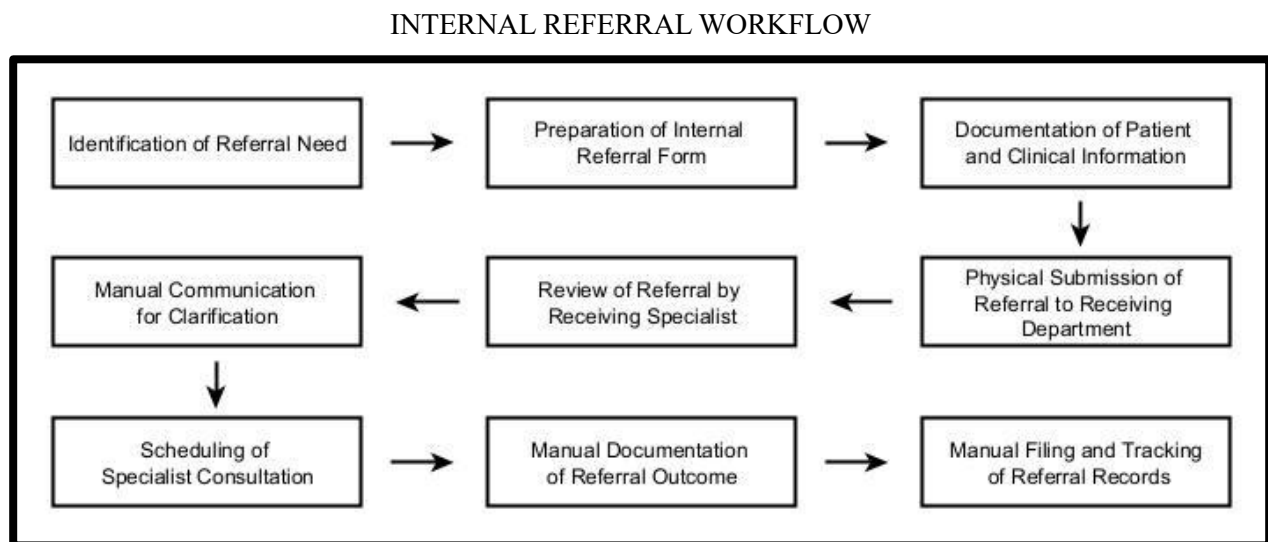


Figure 2.7: Current Internal Referral Workflow

ADMISSION WORKFLOW

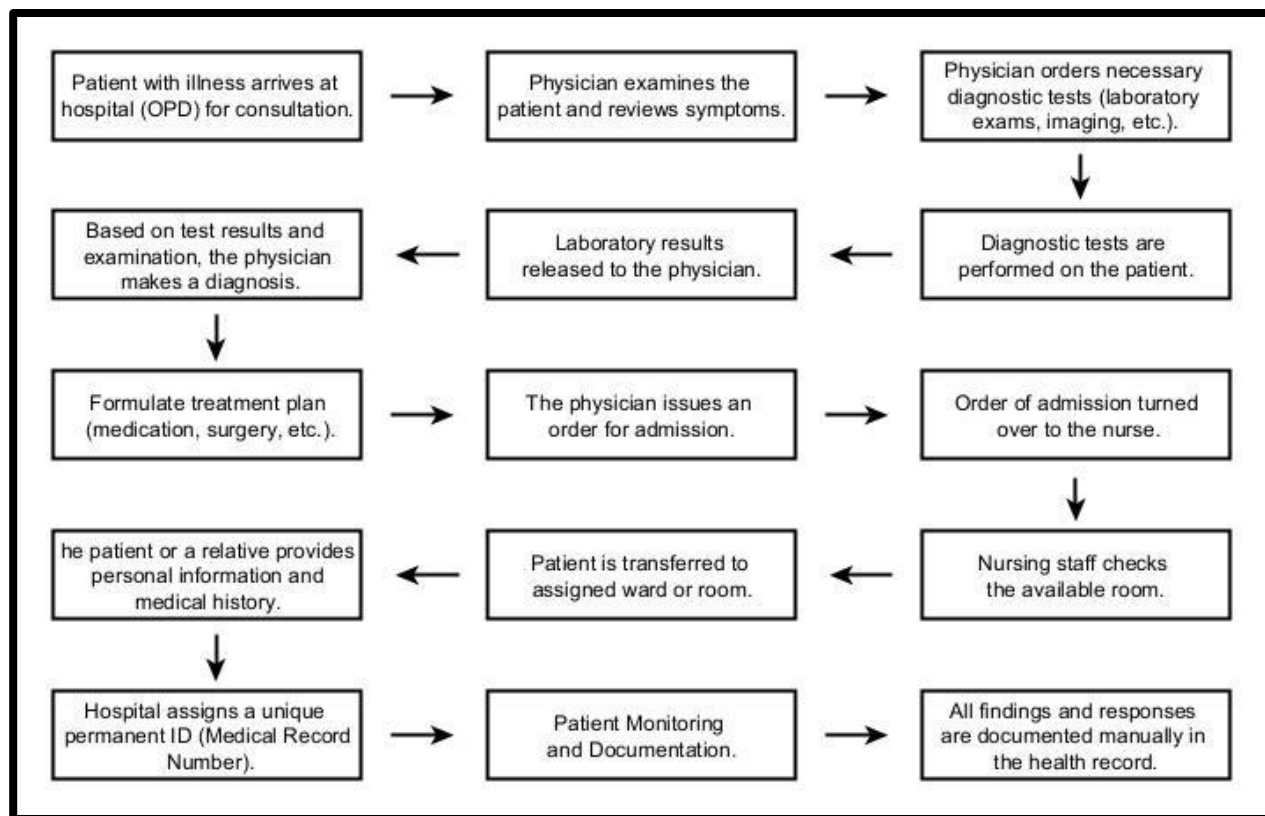


Figure 2.8: Current Admission Workflow

2.5.2 Proposed Workflow

Salaam Hospital's proposed system will manage patient admissions, medical records, and internal referrals through a computerized and centralized digital platform. Patient information will be recorded electronically upon admission and stored securely in a centralized database.

Medical staff will be able to access and update patient records in real time, including medical history, treatments, and referral details. Internal referrals will be processed digitally, allowing faster coordination among departments and reducing reliance on paper-based documentation.

REFERRAL WORKFLOW

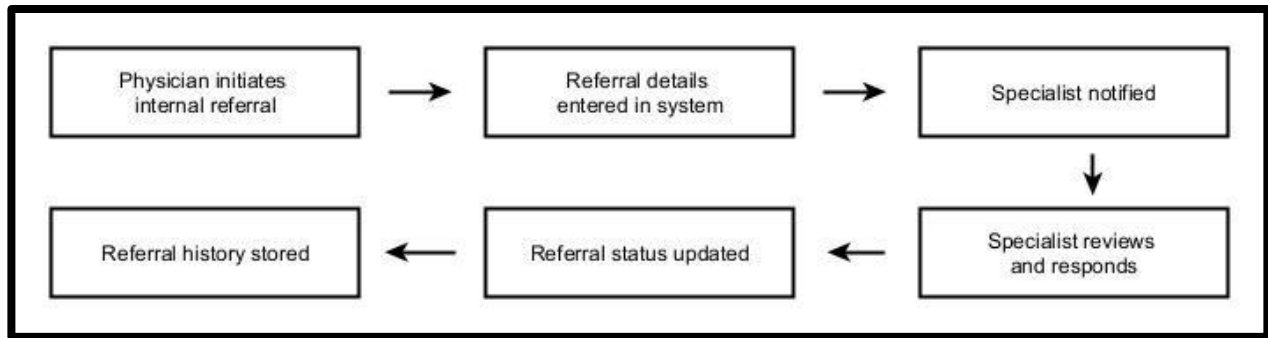


Figure 2.9: Proposed Referral Workflow

ADMISSION WORKFLOW

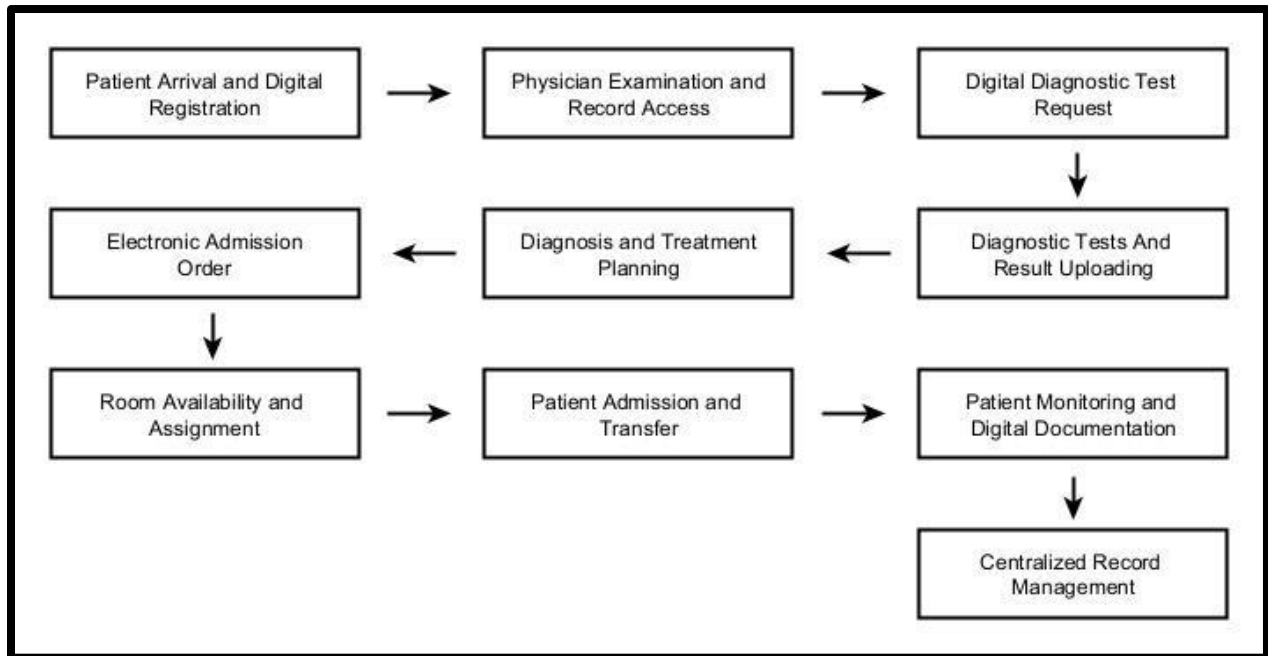


Figure 2.10: Proposed Admission Workflow

Chapter 3

METHODOLOGY

This chapter discusses the methodology of the projects, development phase, results of the conducted tests and the diagrams of the different processes involved in the system. **3.1 Agile Model Methodology**

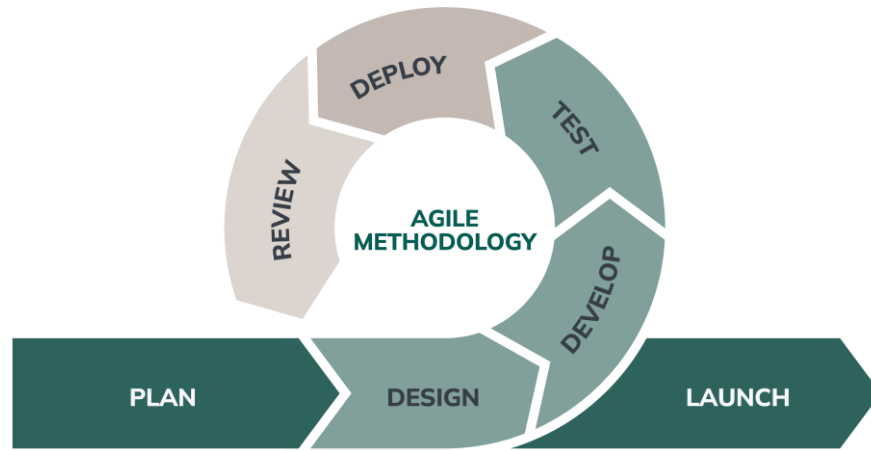


Figure 3.1 Agile Methodology

The developer used Agile Methodology as development life cycle of this project. Agile is a dynamic and iterative method to software development that emphasizes flexibility, teamwork, and response to new demands. This methodology was chosen for this project entitled "Responsive Web-Based Patient Referral and Admission System for Salaam Hospital" because it allows continuous improvement through iterative development cycles.

3.2 Kanban Framework

Kanban was selected as the Agile framework for the development of the Responsive Web-Based Patient Referral and Admission System for Salaam Hospital. This framework emphasizes continuous delivery, visual workflow management, and adaptability, making it highly suitable for a small development team working with changing hospital requirements. Since healthcare systems often evolve due to policy updates and user feedback, Kanban allows tasks to be added, modified, and prioritized without the need for fixed-length iterations, ensuring continuous progress throughout the system development.

The implementation of Kanban in this project followed several core Kanban practices that guided the development process:

1. Visualize the Workflow

A Kanban board was utilized to visually represent all development activities of the system. Tasks were organized into columns such as *To Do*, *In Progress*, and *Done*. This visualization enabled the developers to easily track the status of features such as patient referral processing, admission management, and user interface enhancements, while also providing a clear overview of the project's progress.

2. Limit Work in Progress (WIP)

To avoid task overload and ensure quality output, limits were applied to the number of tasks being worked on simultaneously. This practice helped the development team focus on completing critical modules such as referral approval and patient admission records before starting new tasks, thereby improving efficiency and reducing unfinished work.

3. Manage Flow

The movement of tasks across the Kanban board was continuously monitored to identify delays or bottlenecks in development. This allowed the team to address issues related to system functionality, database integration, or responsiveness early, ensuring a smoother and more efficient development flow.

4. Make Process Policies Explicit

Clear guidelines were established regarding how tasks moved between columns on the Kanban board. For example, a task could only move from *To Do* to *In Progress* once its requirements were clearly defined, and from *In Progress* to *Done* only after successful testing and validation.

This ensured consistency and quality throughout the development process.

5. Implement Feedback Loops

Regular discussions were conducted among the development team to review completed tasks, assess system performance, and address any issues encountered. Feedback from these discussions was used to refine system features, particularly those related to hospital workflows such as patient referrals, admissions, and record management.

6. Improve Collaboratively and Evolve Experimentally

The team continuously improved their workflow by identifying areas for enhancement and adjusting their development practices based on experience and feedback. This collaborative and adaptive approach allowed the system to evolve gradually, ensuring that it effectively meets the operational needs of Salaam Hospital.

3.3 Plan

During the initial phase of the project, the development team conducted a thorough planning process to identify the objectives, scope, and required functionalities of the Responsive Web-Based Patient Referral and Admission System for Salaam Hospital. This phase focused on understanding the existing referral and admission workflow of the hospital and determining how the proposed system could improve efficiency and coordination among hospital staff.

Based on the identified requirements, a project plan was created to organize development activities and monitor progress. The plan outlined the sequence of tasks involved in system analysis, design, development, testing, and deployment. This structured planning approach helped ensure that all necessary activities were properly scheduled and that the project could be completed within the defined timeline.

3.3.1 Requirement Analysis

The requirement analysis phase focused on identifying the functional and non-functional requirements of the system. Requirements were gathered through the review of existing hospital

referral and admission procedures, relevant documentation, and related studies on healthcare information systems. The development team analyzed how patient referrals are currently processed, how admission records are managed, and the challenges encountered in coordinating patient information across departments.

Instead of relying solely on direct interviews, the study utilized documented workflows, system requirements from similar healthcare systems, and existing records provided by Salaam Hospital as reference points. These sources helped define the system requirements, including patient referral submission, admission approval, record management, user access control, and system responsiveness across devices.

The results of the requirement analysis served as the foundation for designing and developing a responsive web-based system that supports efficient patient referrals and admissions while ensuring data accuracy, accessibility, and usability for hospital staff.

3.3.1.1 Empathy Map

Below is a complete empathy map for the user of the Patient Referral and Admission System for Salaam Hospital

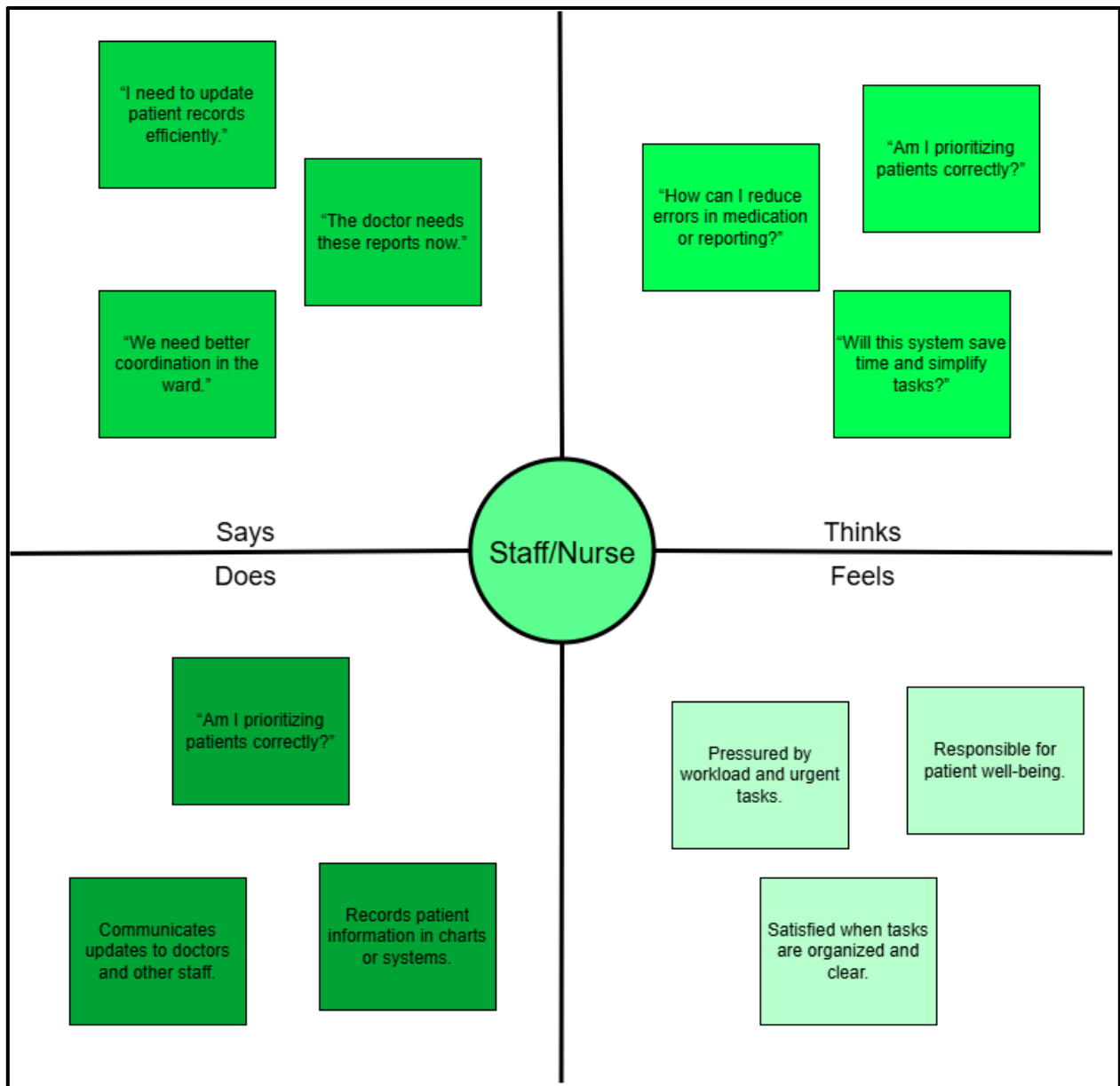


Figure 3.2 Empathy map of the staff/nurse

This empathy map illustrates the daily challenges experienced by hospital staff involved in patient referral and admission processes. By analyzing what staff and nurses think, feel, say, and do during the patient referral and admission process, the proponents gain a deeper understanding of their daily responsibilities, challenges, and workflow pressures. Staff and nurses are often required to manage multiple patients simultaneously while handling documentation, coordination, and

communication tasks, which can become overwhelming when processes are manual and fragmented.

Using the empathy map allows the proponents to design system features that directly address these challenges, such as streamlined referral processing, faster access to patient information, and reduced duplication of data entry. This user-centered approach improves efficiency, minimizes errors, and reduces workload stress among staff and nurses. Moreover, the empathy map serves as a shared reference point that enhances communication among the proponents and guides design decisions throughout system development. Ultimately, it helps validate design choices, prioritize essential functions, and ensure that the Patient Referral and Admission System effectively supports staff and nurses in delivering timely, organized, and quality patient care at Salaam Hospital.

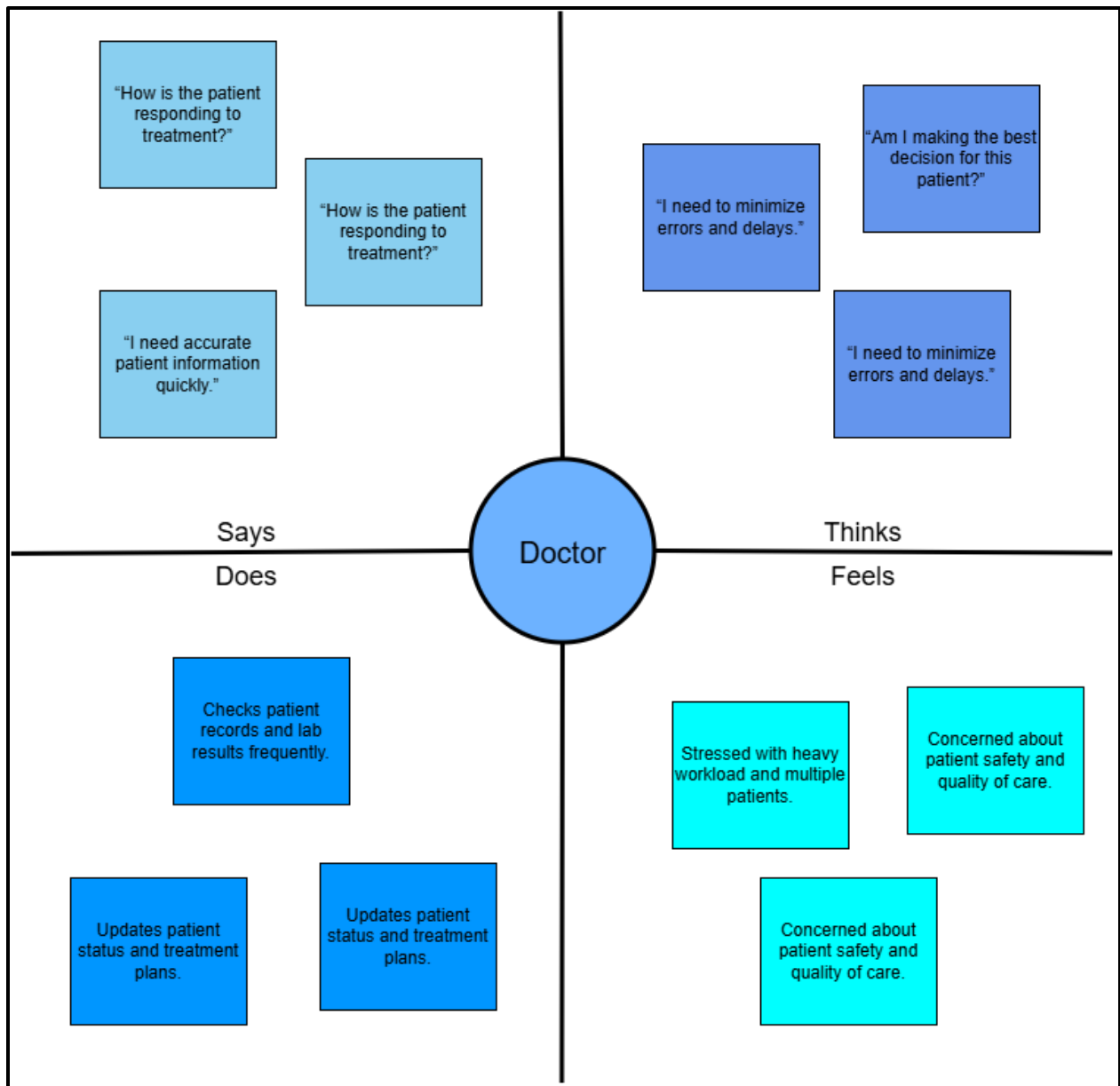


Figure 3.3 Empathy map of the Doctors

By carefully analyzing what doctors think, feel, say, and do during the patient referral and admission process, the proponents gain a deeper understanding of their professional needs, challenges, and expectations. Doctors often require timely access to accurate patient information to make informed clinical decisions, and delays or incomplete records can negatively affect patient care.

Using the empathy map allows the proponents to design system features that directly address real concerns of doctors, such as efficient referral review, quick access to patient medical details, and clear admission status updates. Rather than relying on assumptions, this approach improves the relevance, usability, and clinical value of the system. Additionally, the empathy map serves as a shared reference among the proponents, strengthening communication and guiding design decisions throughout the development process. Ultimately, it helps validate system design choices, prioritize essential functionalities, and identify opportunities to enhance workflow efficiency, making the Patient Referral and Admission System more effective and supportive of doctors' clinical responsibilities.

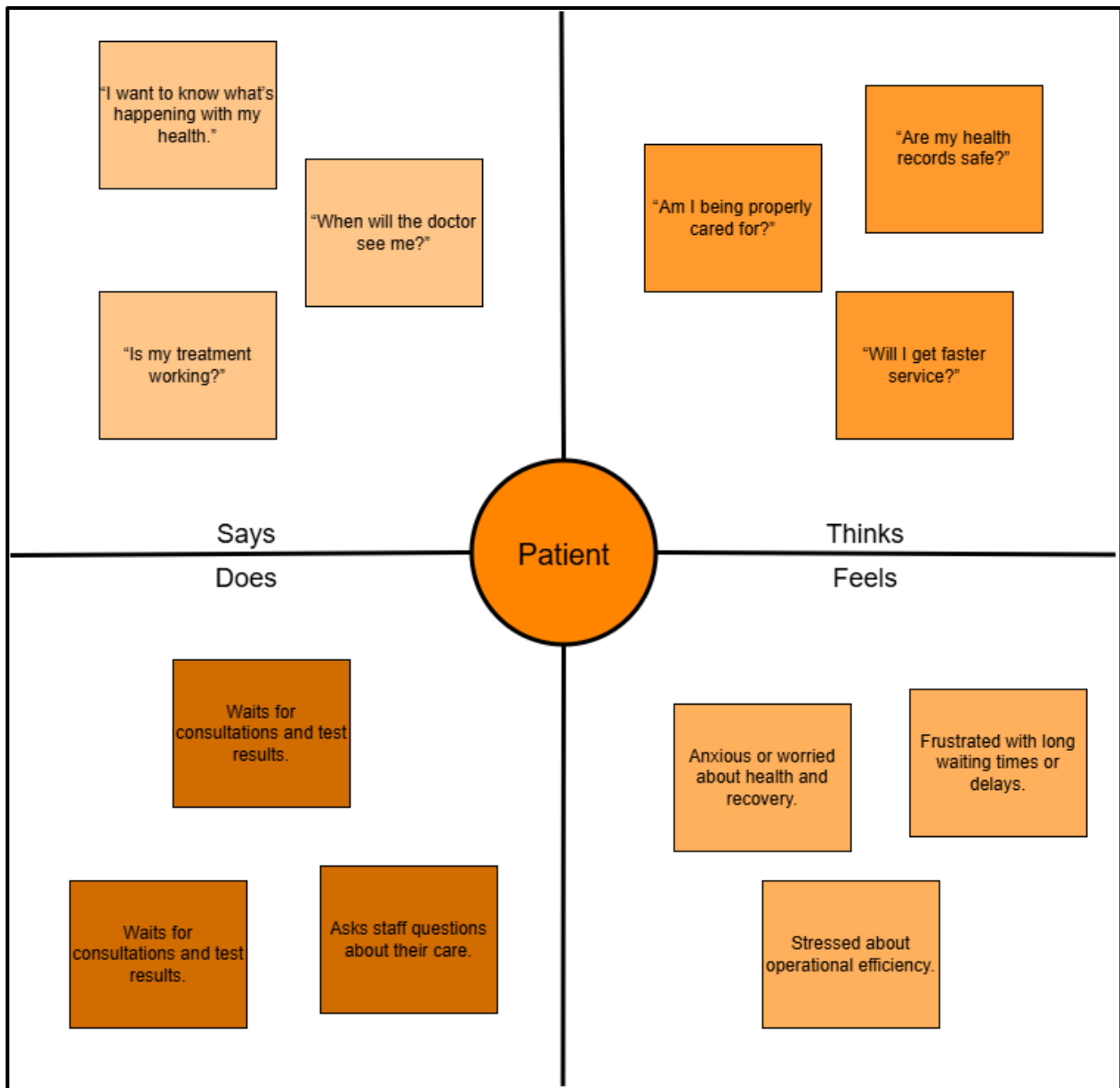


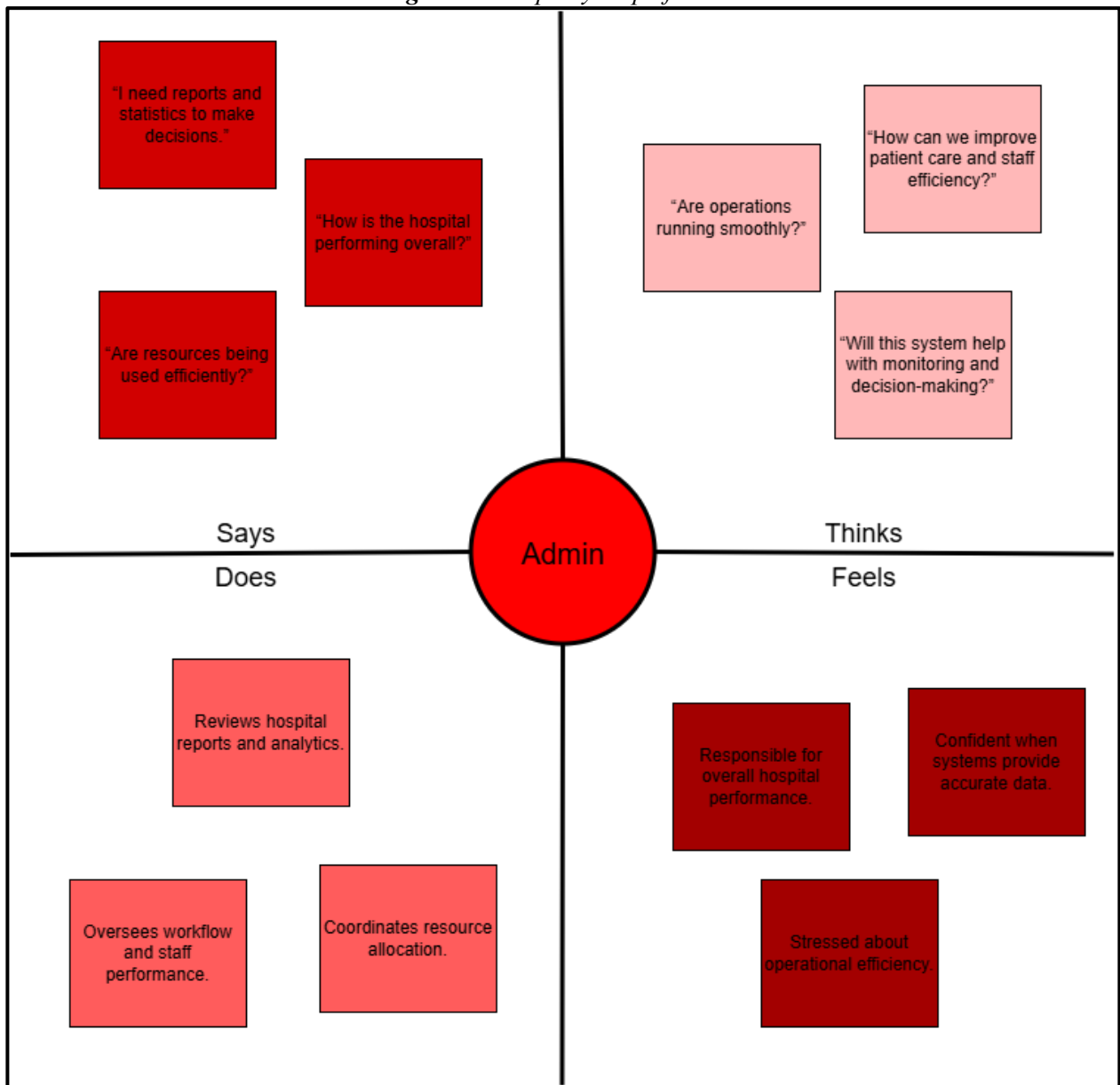
Figure 3.4 Empathy map of the Patient

By examining what patients think, feel, say, and do during the referral and admission process, the proponents gain a clearer understanding of the concerns, expectations, and emotional challenges patients face. Patients often experience anxiety and uncertainty while waiting for referral approvals or admission decisions, especially when communication is delayed or unclear.

Through the use of the empathy map, the proponents are able to design system features that address patient needs such as clear referral status updates, reduced waiting time, and improved transparency

in the admission process. This approach ensures that the system responds to real patient concerns rather than assumptions, enhancing trust, accessibility, and overall user satisfaction. Furthermore, the empathy map serves as a guide in prioritizing patient-centered functionalities and improving communication between patients and hospital staff. Ultimately, this tool supports the development of a more efficient, responsive, and compassionate patient referral and admission system that improves the overall healthcare experience for patients at Salaam Hospital.

Figure 3.5 Empathy map of Admin



By examining what hospital administrators think, feel, say, and do during the management of referrals, admissions, and hospital operations, the proponents gain a clearer understanding of the responsibilities, pressures, and decision-making challenges faced by administrators. Administrators often deal with high workloads, system inefficiencies, and the need to ensure that hospital policies, resources, and patient flow are properly managed. Delays, incomplete data, and lack of real-time system visibility can lead to stress and operational bottlenecks.

Through the use of the empathy map, the proponents are able to design system features that support administrative needs such as centralized data management, real-time monitoring of referral and admission statuses, accurate reporting, and streamlined coordination among departments. This approach ensures that the system assists administrators in making informed decisions, improving efficiency, and maintaining compliance with hospital standards. Furthermore, the empathy map helps prioritize administrative functionalities that enhance oversight, accountability, and communication across the hospital. Ultimately, this tool supports the development of a more organized, efficient, and data-driven patient referral and admission system that strengthens hospital operations and service quality at Salaam Hospital.

3.3.1.2 Functional Requirements and Non-Functional Requirements Table

3.1 Functional Requirements and Non-Functional Requirements

REQUIREMENTS	TYPE	PRIORITY
Centralized Patient Information Database	Functional	Mandatory
Patient Registration and Admission Management	Functional	Mandatory
Internal Patient Referral Management	Functional	Mandatory

Patient Medical History and Record Tracking	Functional	Mandatory
Role-Based User Management (Admin, Doctor, Nurse, Staff)	Functional	Mandatory
Referral Status Tracking and History Log	Functional	Mandatory
Search and Filter Patient Records	Functional	Mandatory
Admission and Referral Monitoring Dashboard	Functional	Mandatory
Document Upload (Referral Forms, Medical Records)	Functional	Mandatory
Automated Notifications (In-App)	Functional	Mandatory
Reporting and Analytics (Admissions, Referrals, Turnaround Time)	Functional	Mandatory
Mobile-Responsive Web Interface	Functional	Mandatory
Secure Login and Authentication	Functional	Mandatory
System Audit Trail and Activity Logs	Functional	Optional
Data Backup and Recovery Module	Functional	Optional

3.3.1.3 Data Gathering

The data required for the development of the Responsive Web-Based Patient Referral and Admission System for Salaam Hospital were not collected through extensive field surveys or external data collection activities. Instead, the developers gathered relevant information through the review of existing hospital documents, records, and documented workflows related to patient admission and internal referral processes.

Data sources included hospital admission forms, referral slips, patient record templates, and existing documentation used by Salaam Hospital staff in their daily operations. These materials were analyzed to identify the data elements required for patient registration, admission tracking, referral processing, and record management. Additionally, the developers reviewed related literature, existing hospital information systems, and best practices in healthcare management systems to ensure that the proposed system aligns with standard hospital workflows and operational requirements.

Rather than conducting separate large-scale interviews or surveys, the developers relied on documented procedures and operational practices currently implemented at Salaam Hospital. This approach ensured that the system requirements were grounded in actual hospital processes and user needs. The collected data served as the basis for defining system features, designing database structures, and developing workflows that accurately reflect the hospital's admission and referral operations.

3.3.2 Gantt Chart

The Gantt Chart presents the project timeline, showing the sequence and duration of the major tasks involved in developing the "Responsive Web-Based Patient Referral and Admission System for Salaam Hospital". It outlines the activities from requirement gathering, design, development, and testing up to documentation. This chart provides a clear overview of the project's progress and ensures that each phase is completed within the planned schedule.

	2026						
	Jan. 25 - Jan. 31	Feb 1. - Feb 7	Feb. 8 - Feb. 14	Feb. 15 - Feb 21	Feb. 22 - Feb. 28	Mar. 1 - Mar. 7	Mar. 8 - Mar. 14
Indanan	Set up development environment	Create database schema	Create login interface	Create logout functionality	Implement role-based access control	Connect dashboard to backend	Add backend to patient registration
Datunaim	Create project folder structure	Create core database tables	Implement login backend logic	Design account management interface	Design system dashboard layout	Implement system navigation menu	Implement patient record validation
Macapodi	Configure database connection	Apply table relationships and constraints	Implement user verification	Add backend to account management	Develop dashboard frontend	Create patient registration interface	Create patient record viewing page
	2026						
	Mar. 15 - Mar. 21	Mar. 22 - Mar. 28	Mar. 29 - Apr. 4	Apr. 5 - Apr. 11	Apr. 12 - Apr. 18	Apr. 19 - Apr. 25	Apr. 26 - Apr. 27
Indanan	Implement patient search functionality	Implement visit room selection	Add backend to referral processing	Create admission interface	Assign room and bed to patient	Handle input validation and errors	Perform workflow testing
Datunaim	Create triage input interface	Display triage status and level	Implement referral tracking status	Add backend to admission recording	Implement access restrictions per role	Integrate all system modules	Fix system bugs
Macapodi	Add backend to triage recording	Create referral assignment interface	Display referral history	Implement room availability checking	Implement system activity logging	Perform module testing	Optimize system performance

Table 3.2 Kanban Board

2026													
System Development	Jan W4	Feb W1	Feb W2	Feb W3	Feb W4	Mar W1	Mar W2	Mar W3	Mar W4	Apr W1	Apr W2	Apr W3	Apr W4
Environment & Database Setup													
Authentication & Account Management													
Dashboard & Navigation													
Patient Registration													
Triage & Referral													
Admission & Room Assignment													
Integration, Testing, & Optimization													

Table 3.3 Gantt chart

3.4 Design

3.4.1 Hierarchical Input-Process-Output

Developers use draw.io for the HIPO, it shows the hierarchical input-process-output of the system. This is graphic illustration of the data flow diagram. It shows the flow of the data and logic within the system.

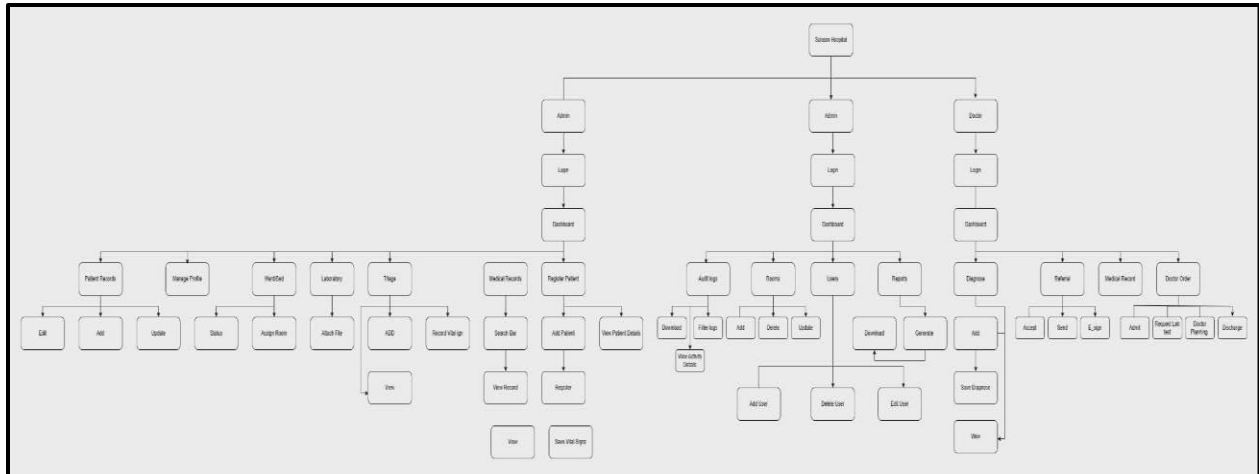


Figure 3.6 Hierarchical Input-Process-Output

3.4.2 Input-Process-Output

3.4.2.1 Patient Registration and Record Management

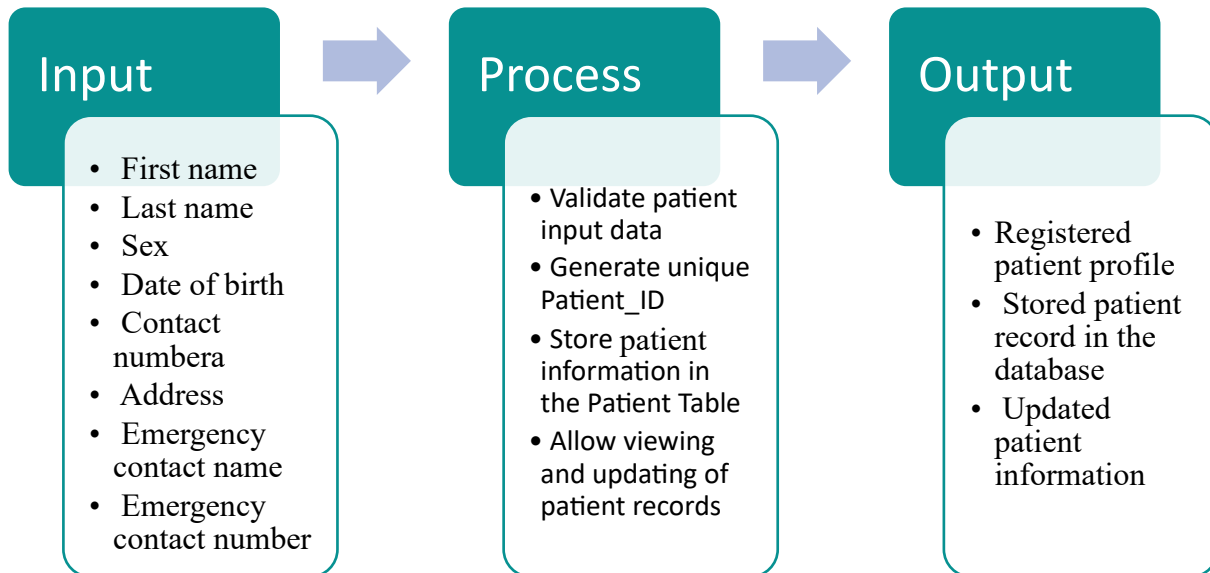


Figure 3.7 Input-Process-Output for Researcher Sign-Up / Registration

Figure 3.7 shows the process of capturing and managing patient information within the system. Patient personal details such as name, sex, date of birth, contact information, and emergency contact details are entered into the system as inputs. The system validates the entered

data to ensure completeness and correctness before generating a unique Patient_ID. Once validated, the information is stored in the Patient table of the database. The system also allows authorized hospital staff to view and update patient records when necessary. As a result, a registered patient profile is created, patient records are securely stored in the database, and updated patient information can be retrieved for future visits.

3.4.2.2 Patient Triage and Internal Referral Management

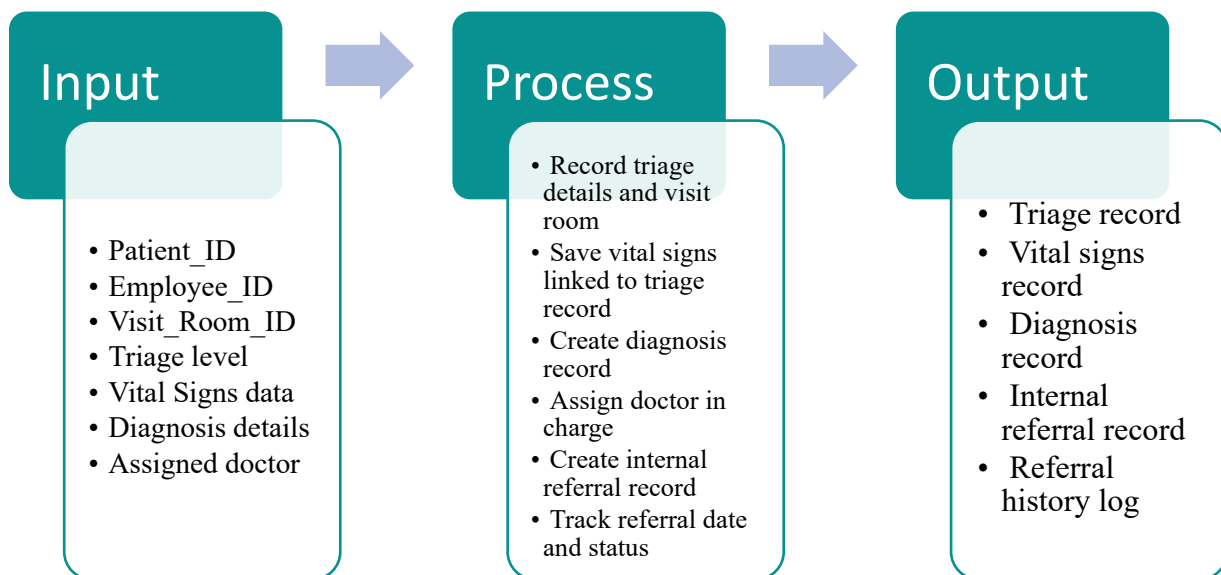


Figure 3.8 Patient Triage and Internal Referral Management

Figure 3.8 shows the workflow for patient triage and internal referral within the hospital. The process begins with inputs such as the Patient_ID, employee details, visit room information, triage level, vital signs, diagnosis details, and assigned doctor. The system records triage data and links the patient to the appropriate visit room. Vital signs are saved and associated with the triage record, after which a diagnosis is created by the attending medical professional. The system assigns a doctor in charge and generates an internal referral record, allowing the referral status and date to

be tracked. The outputs of this process include stored triage records, vital signs records, diagnosis information, internal referral records, and a complete referral history for monitoring patient flow.

3.4.2.3 Patient Admission and Room Assignment

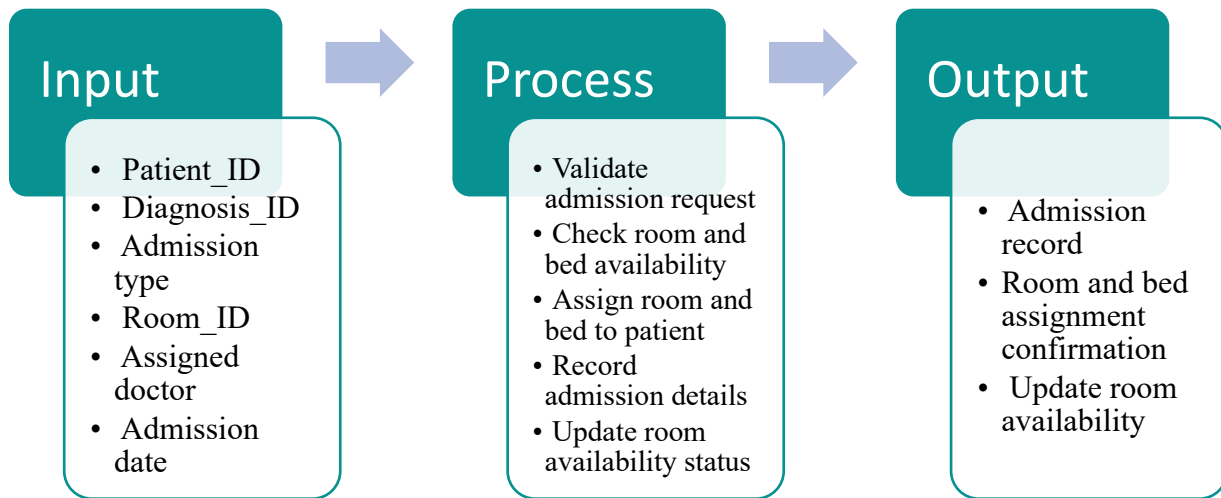


Figure 3.9 Input-Process-Output for System Homepage

Figure 3.9 shows the admission process and room assignment for patients requiring hospitalization. Inputs such as the Patient_ID, diagnosis reference, admission type, room details, assigned doctor, and admission date are submitted to the system. The system validates the admission request and checks the availability of rooms and beds before assigning the appropriate room to the patient. Admission details are then recorded in the database, and room availability status is updated accordingly. The outputs of this process include a confirmed admission record, room and bed assignment details, and an updated room availability record, ensuring accurate monitoring of hospital resources.

3.4.3 Use Case Diagram

Figure 3.18 shows the use case diagram of the "Responsive Web-Based Patient Referral and Admission System for Salaam Hospital". It illustrates how different users interact with the medical appointment system. The system supports four main actors: Nurse/staff, Doctor, and Administrator, each with specific roles and responsibilities.

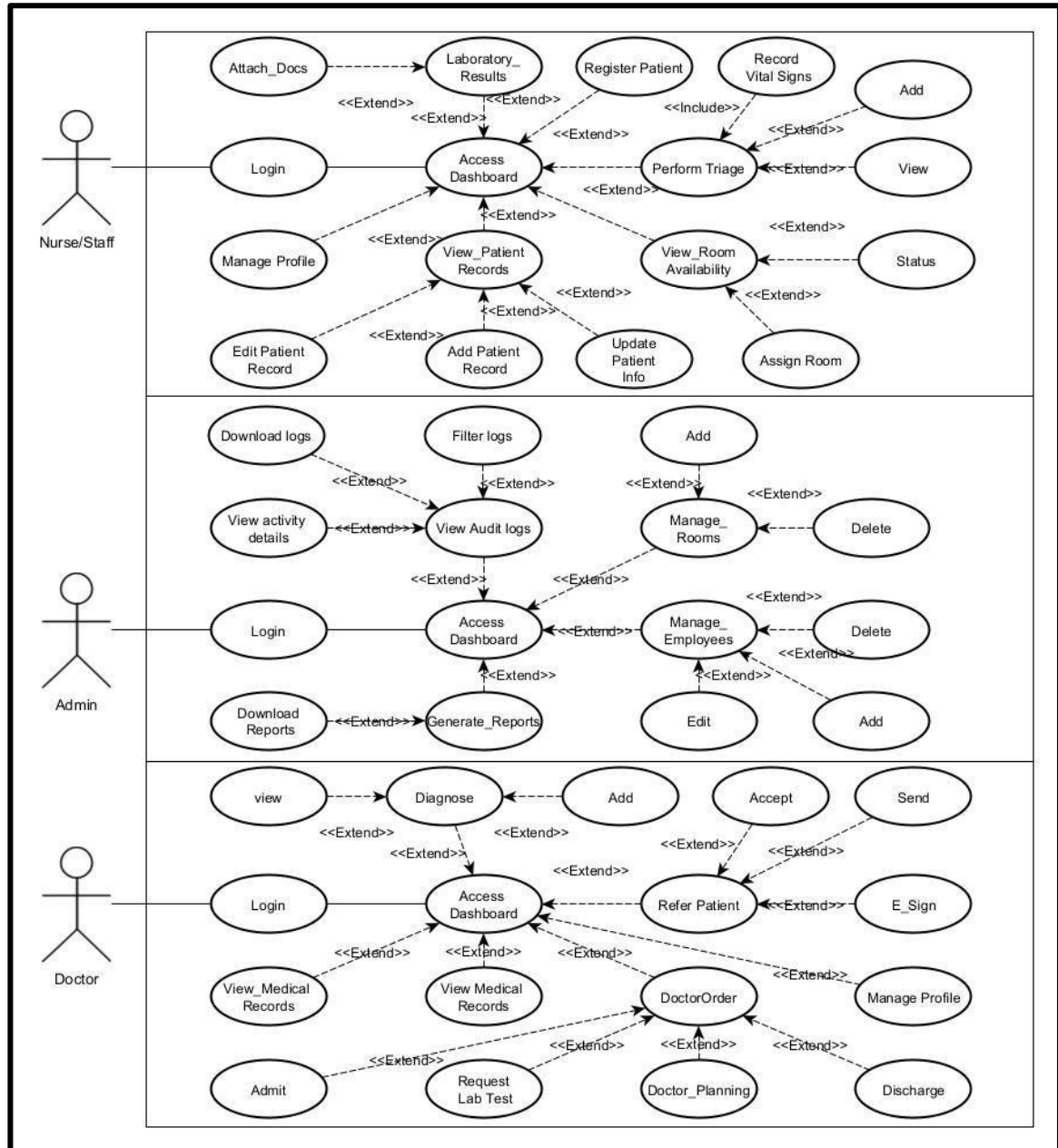


Figure 3.10 Use case Diagram

3.4.4 Entity-Relationship Diagram

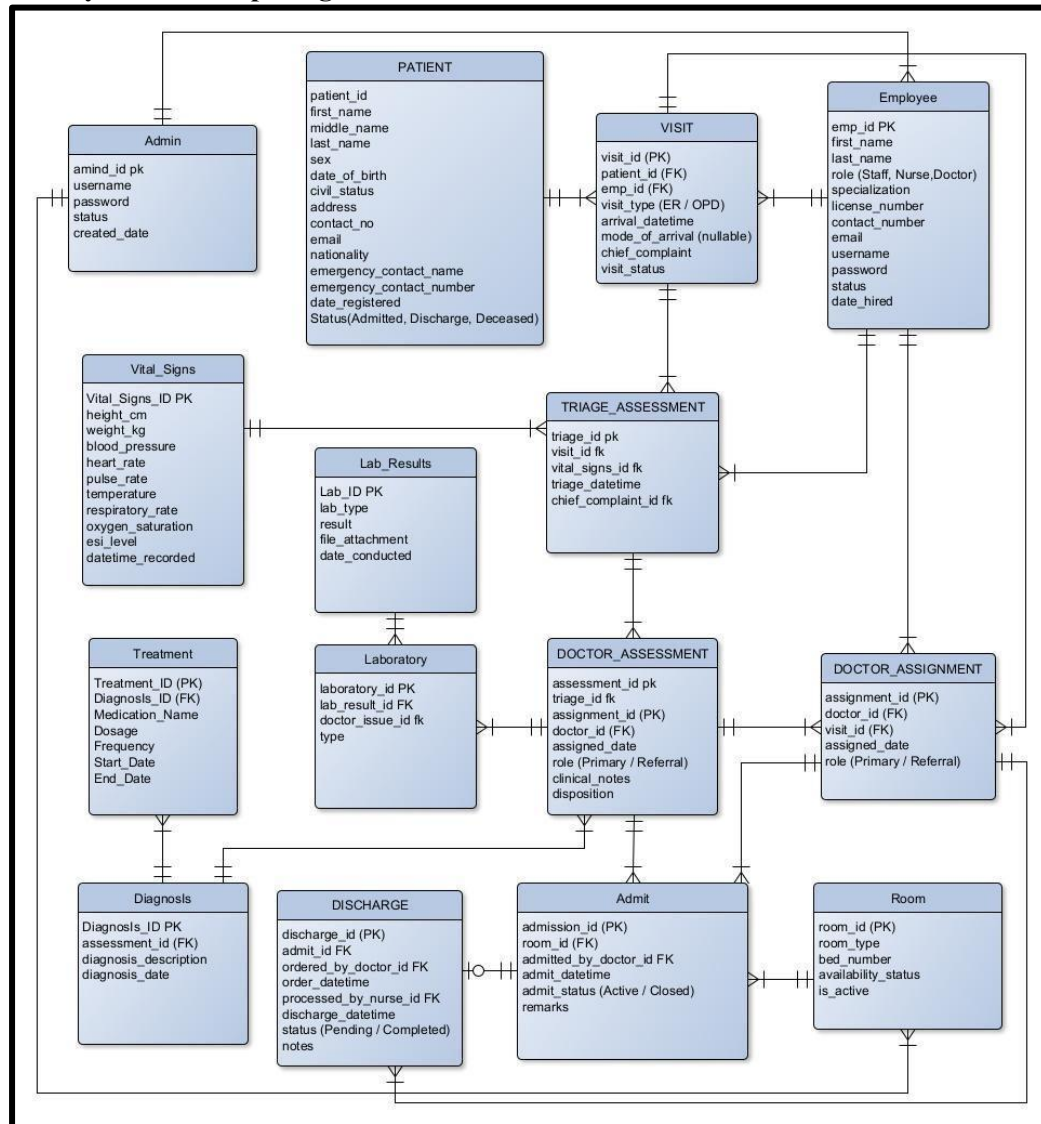


Figure 3.11 Patient Referral and Admission System Entity-Relationship Diagram

Figure 3.11 presents the Entity-Relationship Diagram (ERD) of the "Responsive WebBased Patient Referral and Admission System for Salaam Hospital", illustrating the logical structure of the system's database and the relationships among its core entities. The ERD demonstrates how patient information, clinical processes, hospital resources, and staff roles are interconnected to support efficient emergency care, referral, and admission workflows.

The Patient entity serves as one of the central components of the system, storing essential patient information such as Patient_ID, name, sex, date of birth, contact details, address, and emergency contact information. This entity ensures that patient records are accurately maintained

and can be referenced throughout the triage, diagnosis, laboratory, treatment, referral, and admission processes.

The Employee and Doctor entities represent hospital personnel involved in patient care and system operations. The Employee entity includes staff such as nurses and other ER personnel, while the Doctor entity stores information about licensed physicians, including specialization, contact details, and employment status. These entities support role-based access and accountability within the system. The Admin entity represents system administrators who manage user accounts and system access, ensuring secure authentication and proper system maintenance.

The Triage entity captures initial patient assessment data upon arrival at the emergency room. It is linked to the Patient, Employee, and Visit_Room entities, allowing the system to record triage level (Critical, Urgent, or Non-Urgent), notes, assigned room, and date and time of assessment. Associated with triage is the Vital_Signs entity, which stores recorded health indicators such as blood pressure, heart rate, temperature, and respiratory rate, providing clinical data necessary for medical decision-making.

Medical evaluation and treatment processes are represented by the Diagnosis, Treatment, and Lab_Results entities. The Diagnosis entity records the identified medical condition, the doctor in charge, laboratory references, and diagnosis date. The Treatment entity is linked to Diagnosis and includes prescribed medications, dosage, frequency, and treatment duration. The Lab_Results entity stores laboratory findings related to the patient, including test type, results, file attachments, and dates conducted, supporting accurate and timely diagnosis.

Hospital resource management is handled through the Room, Visit_Room, and Admit entities. The Room entity stores information about room types, bed numbers, and availability status. The Admit entity links patients to assigned doctors and rooms, recording admission type and admission date. These entities ensure proper bed allocation and monitoring of patient admissions within the hospital.

The Referral/Assign entity manages patient referrals and doctor assignments, linking diagnoses to responsible doctors and documenting referral dates. Additionally, the Doctor_In_Charge entity supports tracking of doctor assignments over time, ensuring continuity of care and clear responsibility for patient management.

Overall, the ERD illustrates a well-structured and integrated database design that supports emergency room operations, from patient arrival and assessment to diagnosis, treatment, referral, and admission. The defined relationships ensure data integrity, efficient information flow, and

coordinated collaboration among hospital staff. This design enables the development of a reliable, secure, and efficient emergency referral and admission system that enhances service delivery and patient care at Salaam Hospital.

3.4.5 Architectural Design

The system architecture of the “Responsive Web-Based Patient Referral and Admission System for Salaam Hospital” provides an overview of how the system’s components work together to support efficient patient referral, triage, admission, and hospital resource management. This section presents the arrangement of system layers and the interaction between users, the web-based application, the server-side processing layer, and the centralized database.

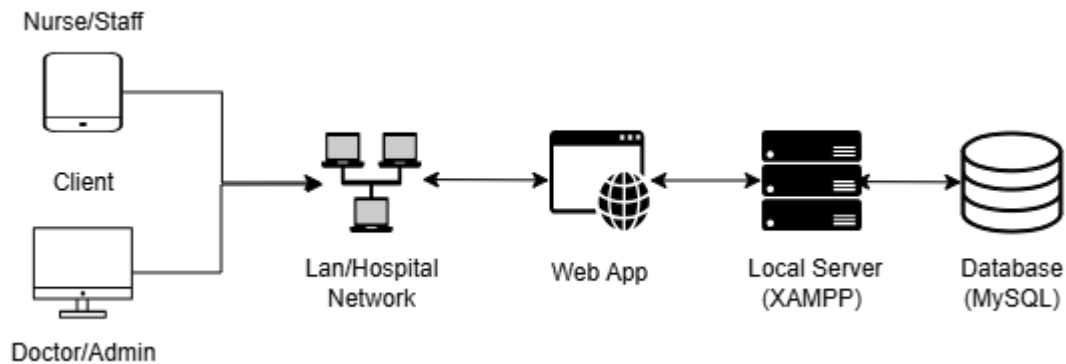


Figure 3.12 Architectural Design

The system follows a layered web-based architecture composed of three main components: the Client Layer, Application Server Layer, and Database Layer. This architectural design ensures clear separation of concerns, efficient data processing, and ease of system maintenance. The Client Layer consists of the web and mobile interfaces used by hospital personnel such as nurses, doctors, and administrators. Through this layer, users can perform tasks such as managing patient records, creating referrals, viewing admission details, and monitoring room availability. The client layer is responsible for presenting information to users and sending user requests to the application server. The Application Server Layer handles the core system logic and business processes. It processes requests received from the client layer, including validating user inputs, managing patient registration, handling referral workflows, assigning doctors, and processing patient admissions.

This layer also enforces system rules and access control before interacting with the database layer. The Database Layer is responsible for securely storing and managing all system data. It contains records related to patients, referrals, diagnoses, admissions, rooms, and user accounts. The application server communicates with the database layer to retrieve, insert, update, and maintain data consistency and integrity. Overall, this architectural design supports efficient referral and admission workflows, secure data handling, and scalable system expansion. By separating the system into distinct layers, the architecture enhances reliability, simplifies future modifications, and ensures smooth operation within the hospital environment.

3.5 Development

3.5.1 Software Specification

Requirements	Details
Operating System	Windows 11
Browser	Google Chrome, Microsoft Edge
Local Server	XAMPP
Database Management System (DBMS)	MySQL
Programming Languages	HTML, CSS, JavaScript
Programming Environment	Visual Studio Code
Framework	React.js
Backend	Node.js

Table 3.4 Developer software requirements

Requirements	Details
Operating System	Windows 10/11, Android, iOS, macOS, Linux
Browser	Google Chrome, Microsoft Edge, Safari, Mozilla Firefox (Latest versions)

Table 3.5 Client software requirements

3.5.2 Hardware Specification

Requirements	Details
Processor	Intel Core i5/Ryzen 5
RAM	8GB – 16GB
Storage	256GB
Operating System	Windows 10/11
GPU	Integrated or dedicated graphics
Peripherals	Mouse, keyboard, monitor

Table 3.6 Developer hardware requirements

Requirements	Details
Processor	Intel Core i3 or equivalent
RAM	4GB
Storage	120GB
GPU	Integrated graphics
Device	Desktop, Laptop, Tablet, or Smartphone
Processor	Intel Core i3 or equivalent

Table 3.7 Client hardware requirements

3.6 Test Plan

The test plan outlines the evaluation process designed to assess the usability, efficiency, and overall functionality of the Responsive Web-Based Patient Referral and Admission System for Salaam Hospital. The primary goal of the testing phase is to ensure that the system is user-friendly, responsive, secure, and capable of supporting accurate patient referral, triage, and admission processes. Testing aims to determine whether the system features meet the operational and informational needs of hospital administrators, doctors, nurses, and staff.

3.6.1 Respondents and Locale

The respondents for the system evaluation will consist of selected hospital personnel and Information Technology students who are familiar with web-based systems. These respondents may include nurses, hospital staff, and BSIT students who can provide both practical and technical feedback. A maximum of 30 respondents will participate in the testing process to ensure a range of user perspectives.

The data gathering and system testing will be conducted at Salaam Hospital and within an academic environment where stable internet access and appropriate devices are available, allowing respondents to interact with the system efficiently and conveniently.

3.6.2 Procedure

The testing procedure will follow a structured and step-by-step approach to collect consistent and measurable feedback. First, respondents will be given a brief orientation and demonstration of the system's features, including login, patient registration, triage recording, referral handling, and admission management.

Second, respondents will be asked to perform specific tasks such as entering patient data, updating referral status, assigning doctors or rooms, and viewing admission records. After completing the assigned tasks, respondents will answer a usability evaluation questionnaire. Qualitative feedback and suggestions will also be collected to identify areas for system improvement.

3.6.3 Tools (Questionnaire)

System Usability Scale (SUS) Questionnaire

Participants rated the following statements using a five-point Likert scale: 1 = Strongly Disagree, 2 = Disagree, 3 = Neutral, 4 = Agree, 5 = Strongly Agree.

Table 3.8 System Usability Scale (SUS) Questionnaire

No.	Statement	1	2	3	4	5
1.	I think that I would like to use this system frequently.					
2.	I found the system unnecessarily complex.					
3.	I thought the system was easy to use.					
4.	I think that I would need the support of a technical person to be able to use this system.					
5.	I found the various functions in this system were well integrated.					
6.	I thought there was too much inconsistency in this system.					
7.	I would imagine that most people would learn to use this system very quickly.					
8.	I found the system very cumbersome to use.					
9.	I felt very confident using the system.					
10.	I needed to learn a lot of things before I could get going with this system.					

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